

MARKETMIND AI

Arabic-first workspace for Egyptian SMEs

Plan · Create · Publish · Improve — with the owner in control

The Problem

Egyptian SMEs often cannot afford expert social-media marketing:

- **Cost** — agencies are out of reach for small businesses
- **Time** — owners already run the operation
- **Skill** — best practice is English, scattered, and hard to translate
- **Trust** — generic AI publishes with no human in the loop

Result: inconsistent posting, wasted budget, no feedback loop.

Target Users

Small and medium businesses in Egypt, across industries:

- Retail · Food and hospitality · Services · Education · Healthcare

The persona is a business owner (not a marketer) who:

- runs one business, one brand,
- has limited budget and time,
- prefers Egyptian Arabic (English available),
- wants practical, explainable recommendations.

The Solution

One guided journey: the system automates research, strategy, and content, while the owner approves every consequential step.

HUMAN APPROVAL REQUIREMENT

If an action can affect the real business, human approval is required.

أي حاجة ممكن تأثر على البيزنس الحقيقي لازم موافقة إنسان واضحة.

Arabic-first (RTL) · explainable and cited · evidence-grounded · owner control · labeled simulation.

Five AI Roles + One Publishing Service

Role	Job	Owner approves?
Discovery	Understand the business → <code>BusinessProfile</code>	Yes
Research	Trusted, cited evidence → <code>ResearchPack</code>	Sources visible
Strategy	12-week marketing plan → <code>StrategyPlan</code>	Yes
Content	Rolling weekly content → <code>ContentPack</code>	Yes, per week
Optimization	Future improvements → <code>OptimizationProposal</code>	Yes, one draft

Publishing is not an AI agent; it is a deterministic, approved action.

The Journey (7 phases)

1. **Discover** — bilingual interview → confirmed profile
2. **Research** — curated RAG + limited web, cited
3. **Strategize** — 4-week plan + 12-week overview + KPIs
4. **Create** — 3-5 posts/week, AR/EN, images, video scripts
5. **Publish/Export** — real n8n publish · export · labeled simulation
6. **Monitor** — read-only Facebook Insights (24h/72h/7d)
7. **Improve** — owner-approved hook/CTA suggestion

Architecture Summary

Monorepo, three deployable apps:

```
apps/web      Next.js 16 (app router, next-intl, RTL, shadcn)
apps/api      NestJS (Prisma + PostgreSQL, Redis/BullMQ workers)
services/ai   FastAPI (LLM providers, RAG, image gen)
packages/contracts shared TypeScript schemas
infra         Docker Compose (Postgres, Redis, Qdrant, n8n)
```

- API ↔ AI over HTTP only (no source imports)
- Postgres = source of truth · Qdrant = rebuildable vector index
- Publishing → n8n workflow (deterministic)

AI and RAG Highlights

- **Curated RAG** — reviewed Arabic/English marketing knowledge in Qdrant (Gemini embeddings, MMR retrieval with lambda 0.5)
- **Profile never embedded** — Business Profile stays in Postgres, never in the shared vector index
- **Provider-neutral** — mock / Gemini / OpenRouter adapters; separate image provider
- **Honesty by design** — every output shows citations, assumptions, knowledge gaps, and blockers

Owner-in-Control Gates

Every consequential step is gated on explicit approval:

- Profile confirmation → before strategy
- Strategy approve/edit/reject → before content
- Content approval → before publish/export
- **Publish approval** → exact candidate + target + mode + schedule
- Optimization approval → one future draft only

The AI never publishes, changes direction, or rewrites history on its own.

Testing and Quality

- **Contracts** — shared schemas validated across web, API, AI
- **API** — unit + e2e (NestJS + Prisma)
- **AI** — pytest for discovery/strategy/content/orchestration, provider contract tests, RAG MMR + retrieval evaluation
- **Web** — Vitest components + Playwright e2e + dictionary parity (`ar` / `en`)
- **Knowledge** — schema + source-resolution validation (`check:marketing-knowledge`)
- One command: `npm run check`

Limitations

- One owner / one business (MVP)
- Facebook publishing only; Instagram/TikTok and paid ads deferred
- Live publishing depends on an approved n8n/Meta integration; otherwise labeled simulation
- Facebook-only monitoring; no manual analytics
- Full video generation, agency dashboards, billing: future work

Roadmap

1. **Commercial readiness** — prepaid points-wallet billing (Egypt gateways)
2. **Broader publishing** — Instagram, scheduling
3. **Deeper optimization** — richer performance signals
4. **Multi-business / agency** — team and account management

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Thank you

Live demo: Discovery → Strategy → Content → review → export/publish